

Q1 Use the city map given in Q1 of assignment 1 and implement the greedy best first search and A* algorithm to reach Boston city from Chicago and compare your results in terms of the number of cities explored before reaching your destination. Consider the heuristic values as given below,

City	h(n) to Boston (miles)
Boston	0
Providence	50
Portland	107
New York	215
Philadelphia	270
Baltimore	360
Syracuse	260
Buffalo	400
Pittsburgh	470
Cleveland	550
Columbus	640
Detroit	610
Indianapolis	780
Chicago	860

Q2 Consider a maze as a 5x5 matrix input where 0 denotes an empty tile, 1 denotes an obstruction/wall, 2 denotes the start state and 3 denotes the reward. Assume valid actions as L,R,U,D where L=move_left, R=move_right, U=move_up, D=move_down,. Use the A* algorithm on the resultant maze for your agent to reach all the rewards, and keep a record of the tiles visited on the way. Consider the rewards as goals. If all goals are completed, then one can come out of the maze. Assume heuristic cost and evaluation cost with proper justification.

Starting Position

